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# Unique integer solution to a system of linear equations via semidefinite programming

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## Abstract

This paper investigates the problem of finding a unique integer solution in  $\{-1, 1\}^n$  to a system of linear equations  $Ax = d$ . Two main approaches are analyzed: a Linear Programming (LP) relaxation and a Semidefinite Programming (SDP) relaxation. For the LP approach, we review existing work that provides necessary and sufficient conditions for the recovery of a unique integer solution and exact expressions for the probability of recovering a unique integer solution, given that  $A \in \mathbb{R}^{m \times n}$  is a generic random matrix. Theoretical considerations prove sharp phase transition at a measurement ratio of  $m/n = 1/2$  and we reproduce the experiments of the initial paper that support this.

A parallel analysis is developed for a proposed SDP relaxation. Analogously to the LP-approach we establish necessary and sufficient conditions for the SDP to recover a unique rank-one solution, which corresponds to the integer solution. We provide a weaker theoretical analysis compared to the LP-approach and derive that recovery of unique integer solutions via the SDP-approach follows if  $m/n \in (1/\sqrt{2}, 1)$ .

Numerical simulations for both LP and SDP approaches are presented, using random Gaussian matrices. The experiments confirm the theoretical threshold of  $\gamma = 1/2$  for the LP method. The simulations also suggest that the same critical threshold of  $1/2$  applies to the SDP method.

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# 1 Introduction

The problem to be considered in this paper is the system of linear equations

$$Ax = d, \quad (1.1)$$

where  $A$  is a given real matrix in  $\mathbb{R}^{m \times n}$ ,  $d \in \mathbb{R}^m$  and  $x \in \mathbb{R}^n$ . More specifically we will consider the instance where  $x$  is an integer solution

$$Ax = d, \quad x \in \{-1, 1\}^n. \quad (1.2)$$

Given  $A$  and  $d$ , finding a unique solution  $x \in \{-1, 1\}^n$  to the equation  $Ax = d$  can be considered as a generalization of the classical knapsack feasibility problem [3], which is an NP-hard problem [9]. In [5] a linear program approach is considered and necessary and sufficient conditions under which (1.1) has a unique solution in  $[-1, 1]^n$ , which is integer, are presented. The authors then move to the setting of  $A$  being a random matrix and the present, under distributional conditions on  $A$ , expressions for when the linear program (LP) recovers a unique integer solution. We present their findings in Section (2) and summarize the chain of arguments used. In Section (3) we present a semidefinite program (SDP) that aims to recover unique integer solutions to (1.1) and then present an analogous analysis as for the linear program highlighting similarities and differences in the approaches. In Section (4) we numerically simulate the linear programming and semidefinite programming approach and connect the empirical results to the previous chapters. Finally in Section (5) we conclude and give an outlook.

We briefly comment on notation: When  $x$  is vector valued, then  $x \geq 0$  means that, each real-valued entry  $x_i \geq 0$ , we also adopt this notation for matrices  $W \geq 0$  means  $W_{ij} \geq 0$  for each entry of  $W$ . Analogously  $x > 0$  means  $x_i > 0$  for each entry. For  $n \in \mathbb{N}$  we denote  $\{1, \dots, n\}$  as  $[n]$ . By  $S^n$  we denote the set of symmetric  $n \times n$  real-valued matrices.

## 2 Linear Programming approach

In this section we present the linear programming approach used in [5] and briefly go through the theoretical underpinnings presented.

### 2.1 Uniqueness of solution

A natural way to relax the problem of finding a unique solution to

$$Ax = d, \quad x \in \{-1, 1\}^n \quad (2.1)$$

is to search over the convex hull of  $\{-1, 1\}^n$ , more precisely, we will consider the relaxation to the linear program

$$\min \quad \|x\|_\infty \quad (2.2)$$

$$\text{s.t.} \quad Ax = d, \quad (2.3)$$

$$x \in [-1, 1]^n \quad (2.4)$$

Any solution  $x$  to problem (2.1) is also a solution to the LP (2.2). Assuming that  $x$  is the unique solution to problem (2.1) we are now interested in understanding necessary and sufficient conditions under which  $x$  is also the unique solution to the LP (2.2).  $x$  being the unique solution to problem (2.2) is equivalent to the affine subspace  $x + \ker A$  intersecting the cube  $[-1, 1]^n$  only at the corner  $x$ , see Figure (1). With this in mind, we can present the necessary and sufficient conditions [5, Proposition 2]:

**Proposition 2.1.** *A necessary and sufficient condition that the LP (2.2) has a unique integer solution is that it has a minimum value of 1 with solution  $x \in \{-1, 1\}^n$  such that for the diagonal matrix  $D$  of  $\pm 1$ 's defined as:  $D = \text{diag}(x)$  the system*

$$DA^\top r > 0$$

*has a solution  $r \in \mathbb{R}^m$ .*

*Proof.* From the definition of LP (2.2) it follows that a necessary and sufficient condition for an  $x \in \{-1, 1\}^n$  to be a solution is for the corresponding minimum value of the objective function to be 1. By definition of the constraints  $x$  is the unique solution if and only if  $(x + \ker A) \cap [-1, 1]^n = \{x\}$ :

$$(x + \ker A) \cap [-1, 1]^n = \{x\} \Leftrightarrow \forall v \in \ker A \setminus \{0\} : x + v \notin [-1, 1]^n.$$

Since  $x \in \{-1, 1\}^n$  this is equivalent to all  $v \in \ker A \setminus \{0\}$  differing from  $x$  in sign in at least one entry, since otherwise take  $v \neq 0$  that has the same structure of signs as  $x$ , then for  $\lambda > 0$  sufficiently small one would have  $x - \lambda v \in [-1, 1]^n$ .

Equivalently: for  $D = \text{diag}(x)$

$$ADy = 0, \quad y \geq 0 \text{ and } y \neq 0,$$

has no solution. By Gordan's Theorem of the Alternative [4, Theorem 2.4.5] this is equivalent to

$$DA^\top r > 0$$

having a solution  $r \in \mathbb{R}^m$ . □

Note that the existence of a solution to  $DA^\top r > 0$  is equivalent to the column vectors of  $AD$  lying in the same hemisphere. See Figure (1) for a geometric interpretation.

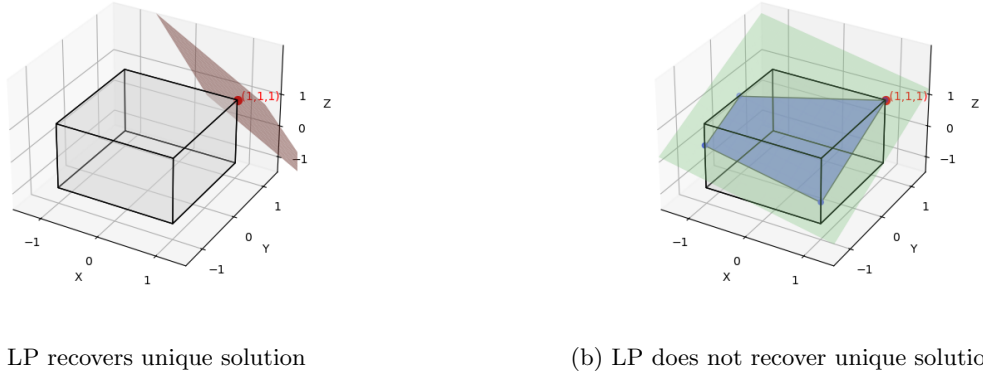


Figure 1: We show examples of hyperplanes corresponding to the case of recovery of the unique solution and the solution space being redundant

## 2.2 Probability of uniqueness

If now  $A$  is a random matrix we are faced with the question of determining the probability of the event that all the columns of  $AD$  lie in the same hemisphere. A random matrix  $A \in \mathbb{R}^{m \times n}$  is called a *generic random matrix* if all all sets of  $m$  columns are linearly independent with probability one and each column vector is symmetrically distributed around the origin. Note that for generic random matrices  $A$  and  $AD$  are distributed identically. Furthermore Wendel [10] uses an inductive argument to derive an exact expression for the probability of the columns of an  $m$  by  $n$  generic random matrix being in the same hemisphere:

**Proposition 2.2.** *For  $m \leq n$ , given random vectors  $a_1, \dots, a_n \in \mathbb{R}^m$  whose joint distribution is symmetric such that any subcollection of  $m$  vectors is linearly independent with probability one, then the probability that the the system*

$$A^\top r > 0 \tag{2.5}$$

*has a solution  $r \in \mathbb{R}^m$  is exactly*

$$p_{m,n} = 2^{-n+1} \sum_{i=0}^{m-1} \binom{n-1}{i}. \tag{2.6}$$

Note that the column vectors  $a_1, \dots, a_n$  satisfy the conditions above by definition, so we can apply Proposition (2.2) onto Proposition (2.1). The expression (2.6) is the same as the probability of observing at most  $m - 1$  heads in a sequence of  $n - 1$  fair coin tosses, so we can use Hoeffding's inequality [2] to upper or lower bound the probability of this event: Let  $\gamma = (m - 1)/(n - 1)$ :

$$\begin{aligned} \gamma > \frac{1}{2} &\Rightarrow p_{m,n} \geq 1 - \exp(-2(\gamma - 1/2)^2(n - 1)) , \\ \gamma < \frac{1}{2} &\Rightarrow p_{m,n} \leq \exp(-2(\gamma - 1/2)^2(n - 1)) . \end{aligned}$$

This predicts that the LP (2.2) will, for fixed  $\gamma > 1/2$  and  $n \rightarrow \infty$ , recover a unique integer solution with probability tending to 1 and analogously for  $\gamma < 1/2$  this probability will tend to 0. Our numerical simulations in Section (4) align with this. We now look at an alternative approach using semidefinite programming.

### 3 Semidefinite Programming approach

In this section we formulate a semidefinite program that aims to solve problem (2.1) and find necessary and sufficient conditions that the semidefinite program recovers a unique integer solution. We then move to the probabilistic setting (generic random matrices) again and try to follow the same chain of thought as for the linear program in deriving expressions for the probability of the SDP recovering a unique integer solution. We highlight differences to the linear program and difficulties that come up.

#### 3.1 Uniqueness of solution

Again we are posed with problem (2.1), namely finding a unique solution to

$$Ax = d, \quad x \in \{-1, 1\}^n .$$

Given that  $x$  is such a solution, multiplying by  $x^\top A^\top = d^\top$  on the right side yields

$$Axx^\top A^\top = dd^\top .$$

Therefore the rank one matrix  $xx^\top$  is a solution to the following feasibility problem (over  $\mathbb{R}^{n \times n}$ )

$$\begin{aligned} \min_X \quad & 0 \\ \text{s.t.} \quad & AXA^\top = dd^\top , \\ & X_{ii} = 1 \quad \text{for } i = 1, \dots, n , \\ & \text{rank}(X) = 1 , \\ & X \succeq 0 . \end{aligned}$$

The rank constraint makes the feasible set non-convex and therefore computationally intractable. Relaxing the rank constraint yields the following semidefinite program

$$\min_X \quad 0 \tag{3.1}$$

$$\text{s.t.} \quad AXA^\top = dd^\top , \tag{3.2}$$

$$X_{ii} = 1 \quad \text{for } i = 1, \dots, n , \tag{3.3}$$

$$X \succeq 0 . \tag{3.4}$$

In a similar approach as with the LP we now try to understand when the SDP (3.1) returns a unique integer solution. We can apply the same chain of thought, we define the operator  $\mathcal{A}$ :

$$\begin{aligned} \mathcal{A}: S^n &\longrightarrow S^m \\ X &\mapsto AXA^\top \end{aligned}$$

and we denote by  $\mathcal{E}_n := \{X \in \mathbb{R}^{n \times n} \mid X \succeq 0, X_{ii} = 1 \text{ for } i = 1, \dots, n\}$  the so called *elliptope*. To prove an analogue proposition to Proposition (2.1) we consider the SDP

$$\min_X \|X\|_1 \quad (3.5)$$

$$\text{s.t. } AXA^\top = dd^\top, \quad (3.6)$$

$$X_{ii} = 1 \quad \text{for } i = 1, \dots, n, \quad (3.7)$$

$$X \succeq 0. \quad (3.8)$$

Note that the SDP (3.5) returns a unique rank one solution if and only if the SDP (3.1) returns a unique rank one solution. Again defining  $D = \text{diag}(x_1, \dots, x_n)$  we can derive necessary and sufficient conditions for recovery of a unique integer solution (see Appendix (A.1) for proof)

**Proposition 3.1.** *A necessary and sufficient condition that the SDP (3.5) recovers a unique rank one solution is that it has a minimum value  $n^2$  with a rank one solution  $xx^\top$  such that*

$$\mathcal{A}(DWD) = 0, \quad \text{s.t. } W \neq 0 \text{ and } 11^\top + W \in \mathcal{E}_n \quad (3.9)$$

*has no solution.*

Note that  $\mathcal{A}(DWD) = ADWDA^\top = (AD)W(AD)^\top$ , so if  $A$  is taken to be a generic random matrix then again the probability that the SDP (3.5) recovers a unique integer solution, and therefore also the SDP (3.1), does not depend on  $x$ , but solely on  $A$ .

### 3.2 Probability of uniqueness

Although Proposition (3.1) for the SDP mirrors Proposition (2.1) for the LP, it seems to be the case that we can not deploy the exact same chain of thought to deduce exact expressions for the probability of unique integer solutions to (3.1) just as we did before, when  $A$  is a generic random matrix. In this section we attempt to use the same argumentation as for the LP and try to show where difficulties come up.

From Proposition (3.1) we can immediately obtain a sufficient condition for SDP (3.1) to recover a unique integer solution (this is not a necessary condition, see Appendix A.2)

**Proposition 3.2.** *If  $x \in \{-1, 1\}^n$  is a solution to  $Ax = d$  and the problem*

$$\mathcal{A}(DWD) = 0 \quad \text{s.t. } W \in S^n \setminus \{0\}, W \geq 0 \text{ and } W_{ii} = 0 \text{ for } i = 1, \dots, n, \quad (3.10)$$

*has no solution, then SDP (3.1) recovers  $xx^\top$  as its unique solution*

*Proof.* The proof follows immediately by noting that

$$W \neq 0 \text{ and } 11^\top + W \in \mathcal{E}_n \Rightarrow W \in S^n \setminus \{0\}, W \leq 0 \text{ and } W_{ii} = 0 \text{ for } i = 1, \dots, n,$$

and then using Proposition (3.1): Note that there exists a solution to

$$\mathcal{A}(DWD) = 0 \quad \text{s.t. } W \in S^n \setminus \{0\}, W \leq 0 \text{ and } W_{ii} = 0 \text{ for } i = 1, \dots, n, \quad (3.11)$$

if and only if there exists a solution to

$$\mathcal{A}(DWD) = 0 \quad \text{s.t. } W \in S^n \setminus \{0\}, W \geq 0 \text{ and } W_{ii} = 0 \text{ for } i = 1, \dots, n, \quad (3.12)$$

by linearity of  $\mathcal{A}$ . □

We state the sufficient condition above, since one can use the same arguments as in Section (2.2) to prove that the SDP (3.1) recovers a unique rank one solution in the regime  $\gamma \in (1/\sqrt{2}, 1)$  with probability tending to one if  $A$  is a Gaussian matrix. For simplicity take<sup>1</sup>  $x = 1$ : since  $\mathcal{A}$  is a linear map from  $S^n$  to  $S^m$ , so therefore the sufficient condition (3.10) is equivalent to a subfamily (since  $W_{ii} = 0$  for  $i = 1, \dots, n$ ) of the column vectors of  $\mathcal{A}$  not having 0 in their convex hull. By Gordan's theorem

<sup>1</sup>Since  $A$  is Gaussian and therefore also a generic random matrix this is no restriction

of the alternative [4, Theorem 2.4.5] this is the equivalent to the subfamily of column vectors lying in the same hemisphere and we can use Wendel’s argument [10]) to give an expression for the probability of this event (see Appendix A.4 for details). It does not seem to be the case that one can use the same line of reasoning to obtain an exact expression for the probability of recovering a unique rank one solution, since in the necessary and sufficient condition (3.9)  $W$  is bounded, since  $11^\top + W \in \mathcal{E}_n$  and it is not clear whether the necessary and sufficient condition (3.9) reduces to whether a finite number of vectors lie in the same hemisphere or not.

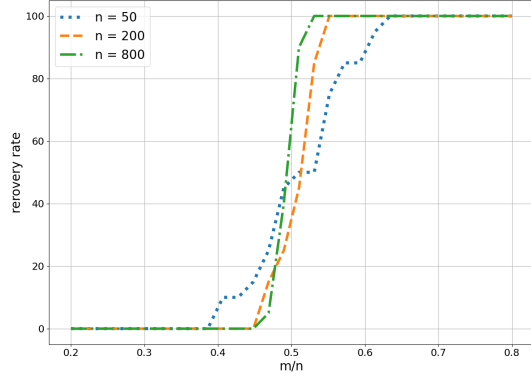
The analysis above is much weaker than the analysis for the LP (2), since we only specify (asymptotically) the behaviour of the SDP (3.1) in the regime  $\gamma \in (1/\sqrt{2}, 1)$  and make no statement about  $\gamma \in (0, 1/\sqrt{2}]$ .

## 4 Numerical Simulations

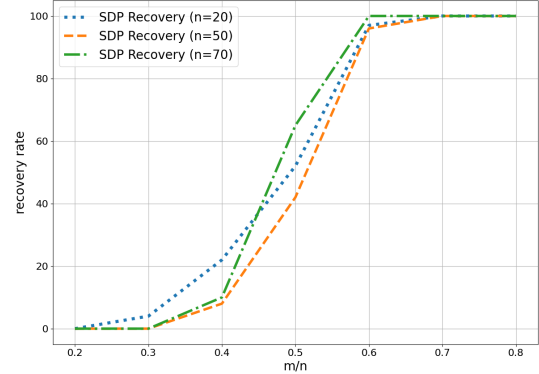
In this section we perform experiments on both the LP (2.2) and the SDP (3.1) and connect our findings to the theoretical considerations in sections (2) and (3):

Taking  $A \in \mathbb{R}^{m \times n}$  to be a random Gaussian matrix, all entries being iid.  $\mathcal{N}(0, 1)$  distributed, implies that  $A$  is generic random matrix. Furthermore we sample  $x$  uniformly from  $\{1, -1\}^n$  and determine  $d \in \mathbb{R}^m$  as  $Ax = d$ . We then can run either the linear program (2.2) or the semidefinite program (3.1) and check whether the returned solution coincides with  $x$ . More specifically:

For the linear program we check if the returned solution has the same signs as the original  $x$  and determine a trial to be successful if this is the case and unsuccessful otherwise. The semidefinite program returns a matrix  $X \in \mathbb{R}^{n \times n}$ . We then project the eigenvector  $v$  corresponding to the largest eigenvalue onto  $x$  (call this  $\text{proj}_x(v)$ ) and check whether  $\text{proj}_x(v)$  or  $-\text{proj}_x(v)$  coincide in signs with the original  $x$ , if either of them align in the signs with  $x$  we determine the trial to be a success and unsuccessful otherwise.



(a) Recovery rate for the Linear Programming approach with 20 trials per each  $(n, \gamma)$ -pair: for  $n = 50, 200, 800$ .  $\gamma$  is taken at 30 equidistant points in  $[0.2, 0.8]$



(b) Recovery rate for the Semidefinite Programming approach with 50 trials per each  $(n, \gamma)$ -pair: for  $n = 20, 50$ . For  $n = 70$  only 20 trials were considered due to computational cost.  $\gamma$  is taken at 7 equidistant points in  $[0.2, 0.8]$

Figure 2: Recovery rates for the linear program (2.2) and the semidefinite program (3.1). The experiments were run using the Python-embedded modelling language CVXPY. For both programs the SCS numerical optimization package was utilized [8].

The numerical simulations for the LP (2.2) confirm the theoretical results and the experiments conducted in [5] and discussed in Section (2), see Figure (2a). The experiments suggest that the same threshold of  $\gamma = 1/2$  seems to hold for the SDP (3.1), see Figure (2b). A comparison of LP (2.2) and the SDP (3.1) on the same problem instances seems to indicate that unique recovery of an integer solution of the LP (2.2) implies unique recovery of a rank one solution of the SDP (3.1), see Appendix (A.3). Note that a proof of this implication would explain the asymptotic behaviour of the SDP (3.1) in the regime  $\gamma \in (1/2, 1)$  and would be a stronger statement than the one proved in Section (3.2) for the regime  $\gamma \in (1/\sqrt{2}, 1)$ .



## 5 Conclusion and outlook

The key finding of this paper is that the proposed SDP (3.1) seems to also come with a critical threshold of  $\gamma = 1/2$  like the LP (2.2) for generic random matrices. It would be of interest to see whether there exists convex relaxation of problem (1.1) that manages to reduce the threshold below  $1/2$  or whether  $1/2$  is a fundamental threshold for which unique recovery of the integer solution becomes computationally intractable, meaning exponential time algorithms have to be deployed. Furthermore comparing LP (2.2) to the SDP (3.1), we observe that the analysis of necessary and sufficient conditions seems to be more subtle for the SDP (3.1). The feasible set of PSD matrices is contained in the elliptope  $\mathcal{E}_n$ , which shows up in various convex relaxations (via SDP) of NP-hard problems (e.g. MaxCut [1, 7]) and while Proposition (3.1) also gives a necessary and sufficient condition for exact recovery of a rank one solution, it is not clear how one can express the exact probability of this event in the setting of generic random matrices, while this is possible in the linear programming case.

## A Appendix

### A.1 Proof of Proposition (3.1)

*Proof.* From the definition of the semidefinite program (3.5) it follows that a necessary condition for a rank one matrix  $xx^\top$  to be a solution is for the corresponding minimum value of the objective function to be  $n^2$ . By definition of the constraints,  $xx^\top$  is the unique solution if and only if  $(xx^\top + \ker \mathcal{A}) \cap \mathcal{E}_n = \{xx^\top\}$ :

$$(x + \ker \mathcal{A}) \cap \mathcal{E}_n = \{xx^\top\} \Leftrightarrow \forall V \in \ker \mathcal{A} \setminus \{0\} : xx^\top + V \notin \mathcal{E}_n. \quad (\text{A.1})$$

We now use the fact that  $\mathcal{E}_n \subset [-1, 1]^{n \times n}$  and since  $xx^\top \in \{-1, 1\}^{n \times n}$ , (A.1) becomes equivalent to:

$$\mathcal{A}(DWD) = 0, \quad \text{s.t.} \quad W \leq 0, W \neq 0 \text{ and } xx^\top + DWD \in \mathcal{E}_n, \quad \text{has no solution}, \quad (\text{A.2})$$

since for  $i, j \in [n]$  one has  $(DWD)_{ij} = W_{ij}x_jx_i$ .  
Note that  $xx^\top = D11^\top D$ , so one has

$$xx^\top + DWD = D(11^\top + W)D$$

and since  $D$  is a self-adjoint isometry and furthermore  $(DWD)_{ii} = W_{ii}$  for *any*  $W \in \mathbb{R}^{n \times n}$  one has

$$D(11^\top + W)D \in \mathcal{E}_n \Leftrightarrow 11^\top + W \in \mathcal{E}_n.$$

This implies that (A.2) is equivalent to:

$$\mathcal{A}(DWD) = 0, \quad \text{s.t.} \quad W \in S^n \setminus \{0\} \text{ and } 11^\top + W \in \mathcal{E}_n \quad \text{has no solution},$$

which yields the claim.  $\square$

### A.2 Counterexample

Consider the counterexample:

$$A = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}$$

Here  $A = 11^\top + W$  with

$$W = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & -1 & 0 \end{pmatrix}$$

The characteristic equation is given by  $\det(A - \lambda I)$ :

$$\det \begin{pmatrix} 1-\lambda & 1 & 1 \\ 1 & 1-\lambda & 0 \\ 1 & 0 & 1-\lambda \end{pmatrix} = (1-\lambda)((1-\lambda)^2 - 2) = 0.$$

However  $\lambda_1 = 1 - \sqrt{2} < 0$  is a root and therefore  $A$  is not PSD and not in  $\mathcal{E}_n$ .

### A.3 Further experiments

We run the linear program (2.2) and the semidefinite program (3.1) on the same instances, meaning: We first sample  $A \in \mathbb{R}^{m \times n}$  with standard normal iid. entries and  $x$  uniformly in  $\{-1, 1\}^n$  and set  $Ax = d$ . Then we run both the LP (2.2) and the SDP (3.1) and again plot recovery percentages. In particular we are interested in whether there are instances in which the SDP (3.1) recovers a solution but the LP (2.2) does not and vice versa, we capture these instances in table (1a). We use the same instances of  $A$  and  $x$  that were used in Figure (2b) and provide a legend: The experiments do not yield any instances in which the LP (2.2) recovers a solution but the SDP (3.1) does not. This seems to indicate that recovery of a unique solution via the (2.2) implies unique recovery via the (3.1). Furthermore every instance that the LP (2.2) solved was also solved by the SDP (3.1) via a rank one

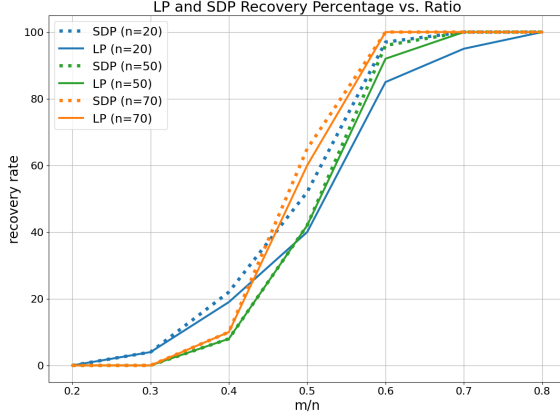


Figure 3: Recovery percentages of LP and SDP: 50 trials per each  $(n, \gamma)$ -pair: for  $n = 20, 50$ . For  $n = 70$  only 20 trials were considered due to computational cost.  $\gamma$  is taken at 7 equidistant points in  $[0.2, 0.8]$ .

Table 1: Recovery analysis

(a) Recovery analysis

| $n$ | Ratio | LP | SDP | diff | R1-diff |
|-----|-------|----|-----|------|---------|
| 20  | 0.40  | 19 | 22  | 3    | 66.67   |
| 20  | 0.50  | 40 | 52  | 12   | 33.33   |
| 20  | 0.60  | 85 | 97  | 12   | 58.33   |
| 20  | 0.70  | 95 | 100 | 5    | 100.00  |
| 50  | 0.60  | 92 | 96  | 4    | 100.00  |
| 70  | 0.50  | 60 | 65  | 5    | 0.00    |

|               |                                                                                                         |
|---------------|---------------------------------------------------------------------------------------------------------|
| $n$           | Problem size (number of variables)                                                                      |
| Ratio         | Measurement ratio $m/n$                                                                                 |
| LP            | Recovery percentage using LP                                                                            |
| SDP           | Recovery percentage using SDP                                                                           |
| difference    | Absolute difference: SDP - LP                                                                           |
| R1-difference | $\frac{\# \text{SDP yields non-rank-one solution, LP does not}}{\# \text{SDP recovers x, LP does not}}$ |

solution. The rank of the returned matrix is calculated by calculating the singular values at a certain threshold and counting the ones that exceed this threshold (here the threshold was at 0.01). We also note that there were instances where the SDP recovered a solution that was rank one and the LP did not recover a solution. Since these instances were rare compared to the total number of trials (1.68%) and further depend on the threshold size, they do not present strong enough evidence for or against the hypothesis that the SDP (3.1) recovers unique rank one solutions with *strictly* higher probability than the LP (2.2). Furthermore there are instances where the SDP (3.1) recovers the true solution via a solution that is not rank 1 and the LP does not (1.26% of trials). We highlight this, although one has to consider threshold sensitivity, these solutions cannot be accounted for by theoretical considerations based on the assumption that the SDP (3.1) recovers the solution if and only if a unique rank one solution is returned.

#### A.4 Recovery of rank one solution in the regime $\gamma \in (1/\sqrt{2}, 1)$

We now prove that for a Gaussian random matrix  $A \in \mathbb{R}^{m \times n}$  and  $\gamma \in (1/\sqrt{2}, 1)$ , where  $\gamma = (m - 1)/(n - 1)$  the probability that SDP (3.1) recovers a unique rank one solution tends to one (given that  $x \in \{-1, 1\}^n$  is a solution to  $Ax = d$ ):

*Proof.* Since we are dealing with a Gaussian random matrix and therefore also with a generic random matrix, we can choose  $x = 1$  without loss of generality, by Proposition (3.1). First note that  $S^n$  is an  $(n + 1)n/2$ -dimensional vector space with basis  $\{e_i e_j^\top + e_j e_i^\top : 1 \leq i \leq j \leq n\}$ . Furthermore any  $W \in S^n \setminus \{0\}$  with  $W_{ii} = 0$  for  $i = 1, \dots, n$  therefore lies in the span of the subfamily of vectors  $\{e_i e_j^\top + e_j e_i^\top : 1 \leq i < j \leq n\}$ . Therefore the sufficient condition (3.10) is equivalent to the vectors  $\{\mathcal{A}(e_i e_j^\top + e_j e_i^\top) : 1 \leq i < j \leq n\}$  not having  $0 \in S^m$  in their convex hull (can be seen by a simple rescaling). By Gordan's theorem of the alternative [4, Theorem 2.4.5] this is equivalent to the vectors  $\{\mathcal{A}(e_i e_j^\top + e_j e_i^\top) : 1 \leq i < j \leq n\} \subset S^m$  lying in the same hemisphere. We can now apply Wendel's argument [10] if the following conditions hold:

1. The joint distribution of the vectors in  $\{\mathcal{A}(e_i e_j^\top + e_j e_i^\top) : 1 \leq i < j \leq n\}$  is symmetric.
2. Any sub-collection of  $m(m + 1)/2$  vectors of  $\{\mathcal{A}(e_i e_j^\top + e_j e_i^\top) : 1 \leq i < j \leq n\}$  is linearly independent in  $S^m$  with probability one.

We first apply Wendel's argument and then argue why the conditions hold:

Since  $S^m$  is  $m(m + 1)/2$  dimensional and the set  $\{\mathcal{A}(e_i e_j^\top + e_j e_i^\top) : 1 \leq i < j \leq n\}$  has cardinality

$n(n-1)/2$  the probability that condition (3.10) holds is exactly<sup>2</sup>

$$p\left(\frac{m(m+1)}{2}, \frac{n(n-1)}{2}\right) = 2^{-\frac{(n-1)n}{2}+1} \sum_{k=0}^{m(m+1)/2} \binom{\frac{(n-1)n}{2}}{k}.$$

Considering the fixed ratio  $\gamma = (m-1)/(n-1)$  and the adjusted ratio  $\gamma' = m(m+1)/(n(n-1))$  one can readily see that  $\gamma' \rightarrow \gamma^2$  and thereby by Hoeffding's inequality [2]

$$\gamma' > \frac{1}{2} \Rightarrow p\left(\frac{m(m+1)}{2}, \frac{n(n-1)}{2}\right) \geq 1 - \exp(-2(\gamma' - 1/2)((n-1)n/2 - 1)).$$

And taking the limit yields

$$\liminf_{n \rightarrow \infty} 1 - \exp(-2(\gamma' - 1/2)(n-1)n/2 - 1) = 1 - \limsup_{n \rightarrow \infty} \exp(-2(\gamma' - 1/2)(n-1)n/2 - 1)$$

which would prove that condition (3.10) holds asymptotically with probability 1, if  $\gamma > 1/\sqrt{2}$  and thereby one would obtain recovery of unique integer solutions with probability tending to one as  $\gamma > 1/\sqrt{2}$  is fixed and  $n \rightarrow \infty$ .

We now argue why the two conditions for Wendel's argument hold:

1. The joint distribution of the vectors in  $\{\mathcal{A}(e_i e_j^\top + e_j e_i^\top) : 1 \leq i < j \leq n\}$  is symmetric.
2. Any sub-collection of  $m(m+1)/2$  vectors of  $\{\mathcal{A}(e_i e_j^\top + e_j e_i^\top) : 1 \leq i < j \leq n\}$  is linearly independent in  $S^m$  with probability one.

Before proving this note that by an elementary calculation we have for  $i, j \in [n]$  that  $\mathcal{A}(e_i e_j^\top + e_i e_j^\top) = a_i a_j^\top + a_j a_i^\top$ , where  $a_i \in \mathbb{R}^m$  denotes the  $i$ -th column vector of  $A$ .

*For 1.:* By symmetry of the Gaussian distribution it immediately follows that the first property holds. Note that this point also holds for a generic random matrix

*For 2.:* Given any subset  $\mathcal{J} \subset \{(i, j) \in [n] : 1 \leq i < j \leq n\}$  such that  $|\mathcal{J}| = (m+1)m/2$ . Since  $\mathcal{A}(e_i e_j^\top + e_i e_j^\top) = a_i a_j^\top + a_j a_i^\top$ , we want to show that with probability one the set  $(a_i a_j^\top + a_j a_i^\top)_{(i, j) \in \mathcal{J}}$  is linearly independent. By rank-nullity theorem this is equivalent to the set  $(a_i a_j^\top + a_j a_i^\top)_{(i, j) \in \mathcal{J}}$  forming a basis of  $S^m$ . This is equivalent to the operator  $\mathcal{A}_{\mathcal{J}}$ , defined as  $\mathcal{A}$  restricted onto  $\text{span}\{e_i e_j^\top + e_i e_j^\top : (i, j) \in \mathcal{J}\}$  being invertible and since  $S^m$  is  $m(m+1)/2$  dimensional, this is the case if and only if the determinant of  $\mathcal{A}_{\mathcal{J}}$  is non-zero. The determinant of  $\mathcal{A}_{\mathcal{J}}$  is a polynomial in the entries of the vectors  $\mathcal{V} = \{a_k : (k, i) \in \mathcal{J} \text{ or } (i, k) \in \mathcal{J} \text{ for some } i \in [n]\}$ . The determinant of a non-zero square matrix is always a non-zero polynomial in the matrix entries and the zero set has Lebesgue measure zero [6]. So the matrices  $(a_i a_j^\top + a_j a_i^\top)_{(i, j) \in \mathcal{J}}$  being linearly dependent is equivalent to the determinant of  $\mathcal{A}$  being zero, which implies that the entries of the vectors of  $\mathcal{V}$  are in a subset of  $\mathbb{R}^{|\mathcal{V}| \cdot m}$  of Lebesgue measure zero, however, since Lebesgue measure (on  $\mathbb{R}^{|\mathcal{V}| \cdot m}$ ) dominates the Gaussian measure (iid.  $\mathcal{N}(0, 1)$ -entries in  $\mathbb{R}^{|\mathcal{V}| \cdot m}$ ) this event has probability zero.  $\square$

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<sup>2</sup>For notational purposes we write  $p_{k,l}$  from equation (2.6) as  $p(k, l)$

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